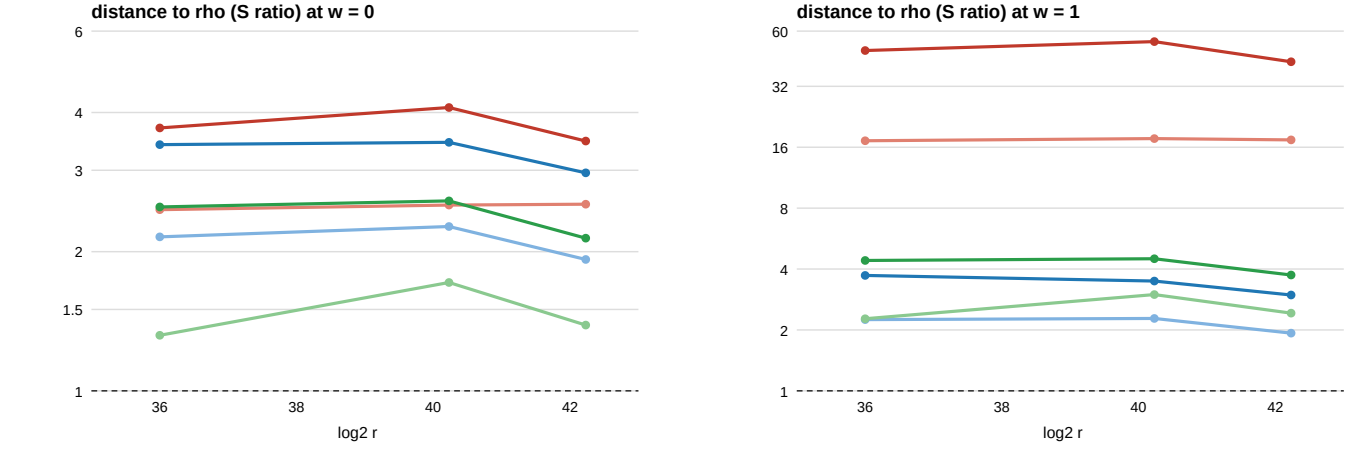
Does the distance to rho move with $\log_2 r$? (priced DP residual collectors)

What was measured. The modes of `research/ic_residual_dp_eq_20261101` (`residual-fused-dp`, `residual-fused-eq`, same per-step pricing) on three prime-order curves: p36 ($2^{36.0}$) from the earlier studies, and p40 ($2^{40.2}$) and p42 ($2^{42.2}$) generated by `scripts/gen_curve.py`. 12 targets × 2 rounds per curve, B = 1 and 4, counted unit, 864 records, 863 verified and one baseline run exhausted (below), `ecbench verify --replay 12` audits OK, A/A = 1.0000. Local L0 run, counts only.

Distance remaining to matched rho (S ratio rho → arm: w = 0 / 0.1 / 1)

arm	p36	p40	p42
res4 (baseline)	3.70 / 11.77 / 48.09	4.10 / 13.04 / 53.28	3.47 / 10.54 / 42.35
res1 (baseline)	2.46 / 5.15 / 17.22	2.52 / 5.27 / 17.65	2.53 / 5.24 / 17.40
res4dp	3.41 / 3.47 / 3.72	3.45 / 3.45 / 3.49	2.96 / 2.97 / 2.98
res1dp	2.15 / 2.17 / 2.25	2.27 / 2.27 / 2.28	1.92 / 1.92 / 1.93
res4eq	2.50 / 2.85 / 4.41	2.58 / 2.92 / 4.50	2.14 / 2.43 / 3.74
res1eq	1.32 / 1.49 / 2.27	1.72 / 1.95 / 2.99	1.39 / 1.58 / 2.42



From 2^{36} to 2^{42} the distance does not grow: res1eq 1.32 → 1.72 → 1.39× rho at w = 0 and 2.27 → 2.99 → 2.42× at w = 1; res1dp 2.15 → 2.27 → 1.92× at w = 0; the baseline res4 3.70 → 4.10 → 3.47× at w = 0 but 48 → 53 → 42× at w = 1. The collector keeps its fixed overhead (√(B + 1) collisions plus per-step priced work) at every size measured; there is no sign of it closing on rho or falling behind with r over this range. Three sizes and 12 targets each cannot fix a slope.

Counted S ratio, candidate / baseline (paired; 95% interval; green = below 1)

curve	pair	w = 0	w = 0.1	w = 1
p40	res1dp/res1	0.898 [0.683, 1.182]	0.430 [0.327, 0.566]	0.129 [0.098, 0.170]
p40	res1eq/res1	0.680 [0.495, 0.935]	0.369 [0.269, 0.508]	0.169 [0.123, 0.233]
p40	res4dp/res4	0.841 [0.742, 0.953]	0.265 [0.234, 0.300]	0.065 [0.058, 0.074]
p40	res4eq/res4	0.628 [0.533, 0.740]	0.224 [0.190, 0.264]	0.084 [0.072, 0.100]
p42	res1dp/res1	0.759 [0.601, 0.959]	0.368 [0.293, 0.461]	0.111 [0.089, 0.138]
p42	res1eq/res1	0.548 [0.444, 0.675]	0.301 [0.247, 0.367]	0.139 [0.115, 0.169]

p42	res4dp/res4	0.438 [0.148, 1.295]	0.285 [0.251, 0.323]	0.071 [0.063, 0.081]
p42	res4eq/res4	0.316 [0.110, 0.914]	0.233 [0.212, 0.255]	0.089 [0.082, 0.097]
p36	res1dp/res1	0.873 [0.695, 1.097]	0.422 [0.336, 0.530]	0.131 [0.104, 0.164]
p36	res1eq/res1	0.535 [0.423, 0.677]	0.290 [0.229, 0.367]	0.132 [0.104, 0.168]
p36	res4dp/res4	0.920 [0.782, 1.083]	0.295 [0.250, 0.346]	0.077 [0.066, 0.091]
p36	res4eq/res4	0.675 [0.595, 0.764]	0.242 [0.213, 0.274]	0.092 [0.081, 0.104]

Frozen rule C1 (p40 and p42, 8 cells, below 1 at w = 0, 0.1 and 1): **not met**, 6 of 8 cells hold. The failures are both at w = 0: p40 res1dp [0.683, 1.182] (12 targets, wide) and p42 res4dp [0.148, 1.295]. The latter comes from one **censored baseline run**: res4 on p42, round 1, was `exhausted` at 10^8 trials (S = 87.9 recorded). The baseline stores every residual in a table capped at 2^{22} entries (`TABLE_CAP`); at 2^{42} it needs about $8 \cdot 10^6$ steps, so after the cap later steps can only collide with the first $4.2 \cdot 10^6$ residuals, collisions become rarer, and this run hit the trial limit. The candidates store only distinguished points and stay far below the cap. All w = 0.1 and w = 1 cells hold (0.22–0.43 and 0.065–0.17).

Classification and caveats

- Engineering of a generic collision search, as before; rho stays ahead at every weight (1.3–3.0× for the best arms), no structural index-calculus content, no claim to beat rho.
- The baseline's table cap is a limit of the existing collector, not of the candidates, and it affects only the baseline arms at large r; the censored pair is reported, not dropped.
- Curves p40 and p42 were generated for this study (random curve with a prime group order, found by a Hasse-interval BSGS); gen_curve.py checks that N is prime, lies in the Hasse interval and that N·P = O for the generator P, which fixes the group order. Distinct seeds from earlier studies, so sessions are not pooled.
- No `VALIDATION.json`; no leaderboard or benchmarks row. Isolation L0; L2 specs committed for an isolab-eks replay.